

Problem 18.21

Victor wants to purchase a perpetuity paying 100 per year with the first payment due at the end of year 11. He can purchase it by either

- (1) paying 90 per year at the end of each year for 10 years, or
- (2) paying K per year at the end of each year for the first 5 years and nothing for the next 5 years.

Calculate K .

Solution

Let v be the annual discount factor. The perpetuity pays 100 per year forever, with the first payment at the end of year 11. Therefore, its present value at time 0 is

$$100a_{\infty|}v^{10}.$$

Using payment option (1), this value must equal the present value of payments of 90 at the end of each year for 10 years:

$$90a_{\overline{10}|} = 100a_{\infty|}v^{10}.$$

Since

$$a_{\overline{10}|} = \frac{1 - v^{10}}{i} \quad \text{and} \quad a_{\infty|} = \frac{1}{i},$$

we get

$$90 \left(\frac{1 - v^{10}}{i} \right) = 100 \left(\frac{1}{i} \right) v^{10}.$$

Canceling $1/i$ gives

$$90(1 - v^{10}) = 100v^{10}.$$

Thus,

$$90 = 10v^{10} \quad \implies \quad v^{10} = \frac{9}{10}.$$

Hence,

$$v^5 = \sqrt{\frac{9}{10}}.$$

Using payment option (2), the present value of the first 5 payments of K must also equal the present value of the 10 payments of 90:

$$Ka_{\overline{5}|} = 90a_{\overline{10}|}.$$

Substitute the annuity formulas:

$$K \left(\frac{1 - v^5}{i} \right) = 90 \left(\frac{1 - v^{10}}{i} \right).$$

Canceling $1/i$,

$$K(1 - v^5) = 90(1 - v^{10}).$$

Since $1 - v^{10} = (1 - v^5)(1 + v^5)$,

$$K(1 - v^5) = 90(1 - v^5)(1 + v^5).$$

Therefore,

$$K = 90(1 + v^5).$$

Substitute $v^5 = \sqrt{9/19}$:

$$K = 90 \left(1 + \sqrt{\frac{9}{19}} \right) = 151.9422481.$$

Thus,

$$\boxed{K = 151.94}.$$